Climate-Feature Dashboard for Semiconductor Technology

1 Project Overview

The Climate-Feature Dashboard for Semiconductor Technology project connects incident response, threat intelligence, cybersecurity, and data integrity with climate literacy.

The goal of the project is to design and implement a secure dashboard that identifies technological features used in semiconductor chips and explains how those features may affect climate change.

Relevant technological and environmental features may include:

Energy consumption, Electricity source and grid carbon intensity, Water consumption, Materials used in semiconductor manufacturing , Semiconductor packaging materials, Manufacturing location, Waste generation etc

The project will use reliable data, documented sources, and transparent assumptions to create an evidence-based dashboard.

A second version of the dashboard will represent an adversarial scenario. In this version, attackers, vendors, malicious insiders, or influence actors may intentionally manipulate data, remove important features, modify assumptions, or redesign visualizations to create a misleading impression that semiconductor production has little or no effect on climate change.

Students will compare the accurate and manipulated dashboards to demonstrate how changes in data provenance, feature selection, assumptions, visualization, or system design can influence climate-related conclusions and user trust.

2 Project Deliverables

The project will include six major deliverables.

Each deliverable should demonstrate an understanding of semiconductor technology, climate impacts, cybersecurity, threat modeling, data integrity, incident response, secure system design, and adversarial manipulation.

2.1 Proposal and Data-Source Plan

The team will develop a project proposal explaining the purpose, scope, research question, and planned structure of the Climate-Feature Dashboard.

The proposal should identify the semiconductor-related technological features that will be studied and explain why those features are important to climate change.

2.1.1 Research Papers

The team will select at least five reliable academic or peer-reviewed research papers related to semiconductor technology and climate or environmental impacts.

The selected papers should identify measurable semiconductor technological features and explain how those features may contribute to climate change or other environmental impacts.

For each research paper, the team should provide:

- Full academic citation
- Research objective
- Semiconductor technology or manufacturing feature studied
- Data or variables used in the study
- Environmental or climate impact discussed
- Important findings
- Limitations of the study
- Explanation of how the paper supports the dashboard

2.1.2 Data-Source Plan

The team must develop a data-source plan explaining what data will be collected and used in the dashboard.

For each dataset, the team should document information such as:

- Dataset or source name
- Source organization
- Publication or reporting year
- Data year
- Measurement unit

- Semiconductor feature represented
- Geographic scope
- Lifecycle boundary
- Reliability of the source
- Limitations
- Method used to validate the data

Possible sources may include:

- Peer-reviewed academic research
- Semiconductor company sustainability reports
- Government databases
- Industry reports
- Environmental datasets
- Electricity-grid datasets
- Water-scarcity datasets
- Lifecycle-assessment studies
- Environmental protection agencies
- International environmental organizations

The proposal should also explain how data from different sources will be validated before being included in the dashboard.

2.2 Threat Model and Data-Integrity Plan

The team will create a cybersecurity threat model for the Climate-Feature Dashboard.

The purpose of the threat model is to identify how attackers, malicious insiders, vendors, or influence actors could manipulate dashboard data and create misleading climate conclusions.

2.2.1 Attacker-versus-Defender Scenario

The project should contain a realistic attacker-versus-defender scenario.

For example, dashboard data may be stored in:

- A web server
- A cloud environment
- A database
- A shared repository
- A CSV file
- A spreadsheet
- An API

The team should identify possible methods an attacker could use to gain unauthorized access or manipulate information.

Possible attacks may include:

- Stealing administrator credentials
- Phishing a dashboard administrator
- Exploiting a vulnerable server
- Exploiting a vulnerable web application
- Modifying CSV files
- Modifying spreadsheets
- Changing database records
- Manipulating API responses
- Changing greenhouse-gas values

2.2.2 Threat-Model Components

The threat model should identify:

- Important system assets
- Potential attackers
- Attacker motivations
- Possible attack vectors
- System vulnerabilities
- Data-integrity risks
- Indicators of compromise
- Potential consequences of successful manipulation

2.2.3 Defender Responsibilities

The team should explain how defenders will protect the dashboard and its data.
Possible security controls include:

- Multi-factor authentication
- Strong password policies
- Role-based access control
- Least-privilege access
- Encryption
- File hashing etc

2.2.4 Incident-Response Plan

The project should apply knowledge from the course about incident response.
The incident-response section should include:

1. Preparation
2. Detection and Analysis
3. Containment

4. Eradication
5. Recovery
6. Lessons Learned

The team should explain how investigators could determine:

- What information was changed
- When the change occurred
- Who may have performed the modification
- Which account was used
- What systems were affected
- Which environmental metrics were affected
- Whether visualization settings were modified
- Whether source information was removed
- Whether assumptions were changed
- How the trusted version of the dashboard can be restored

2.3 Secure Dashboard Prototype

The team will design and build a secure dashboard prototype that presents the selected semiconductor technological features and explains their relationship to climate change.

The dashboard may be developed using an online dashboard platform, spreadsheet visualization tool, web-development framework, Python application, or another approved technology.

Possible tools include:

- Tableau
- Microsoft Power BI
- Google Looker Studio
- Microsoft Excel
- Python Streamlit

- Python Flask
- HTML
- CSS
- JavaScript

2.3.1 Dashboard Visualizations

The dashboard should include multiple visualizations that help users understand the environmental impacts associated with semiconductor technology.

Possible dashboard components include:

- Energy-consumption charts
- Water-use charts
- Greenhouse-gas emission charts
- Semiconductor process-node comparisons
- Packaging-material comparisons
- Manufacturing-location comparisons
- Renewable versus nonrenewable electricity comparisons
- Scope 1 emission comparisons
- Scope 2 emission comparisons
- Scope 3 emission comparisons
- Carbon-intensity indicators
- Water-scarcity indicators
- Lifecycle-impact charts
- Supply-chain maps
- Technology-efficiency comparisons
- Historical trends
- Company comparisons
- Facility comparisons

2.3.2 Dashboard Transparency

The dashboard should clearly identify:

- Data source
- Data year
- Measurement unit
- Geographic location
- Lifecycle boundary
- Assumptions
- Limitations
- Confidence or uncertainty when appropriate

2.3.3 Dashboard Security Features

The secure dashboard should include data-integrity features such as:

- Source tracking
- Version information
- File hashes
- Validation rules
- Audit information
- Data provenance
- Access control
- Change history

The dashboard should allow users to distinguish between:

- Verified information
- Estimated values
- Assumptions
- Calculated or derived values

2.4 Accurate and Manipulated Dashboard Comparison Analysis

The team will create or analyze a second version of the dashboard representing an adversarial or manipulated scenario.

The manipulated dashboard should intentionally demonstrate how an attacker, malicious insider, vendor, or influence actor could change users' interpretations of semiconductor climate impacts.

Possible manipulations include:

- Altering numerical values
- Deleting environmental features
- Hiding unfavorable information
- Changing assumptions
- Using misleading chart scales
- Changing chart colors
- Removing source information
- Selecting only favorable years
- Selecting only favorable companies
- Selecting only favorable facilities
- Changing measurement units
- Hiding geographic information
- Removing water-scarcity information
- Excluding process-gas emissions
- Changing lifecycle boundaries
- Using percentages instead of absolute values
- Replacing absolute emissions with intensity metrics
- Cherry-picking favorable data
- Removing uncertainty information
- Removing limitations
- Changing labels or descriptions to influence users

2.4.1 Comparison Requirements

The team will compare the original secure dashboard with the manipulated dashboard.

The comparison should identify:

- What the original feature or value was
- What was changed
- How the manipulation was performed
- Why the manipulation may be difficult for users to notice
- How the manipulation changes the climate-related conclusion
- How the manipulation affects user trust
- Which security control could detect the manipulation
- Which security control could prevent the manipulation
- What evidence an incident responder could use during an investigation

The team should include screenshots or side-by-side visualizations showing the accurate and manipulated dashboards.

The analysis should demonstrate that data manipulation does not always require changing a numerical value.

An attacker may manipulate conclusions by:

- Removing important variables
- Changing assumptions
- Changing visualization design
- Hiding data provenance
- Changing lifecycle boundaries
- Selecting favorable metrics
- Removing uncertainty

2.5 Final Written Report

The team will submit a final written report documenting the complete project.

The report should explain the connection between semiconductor technology, climate change, data integrity, cybersecurity, threat intelligence, and incident response.

The final report should include the following sections:

1. Title Page
2. Abstract or Executive Summary
3. Introduction
4. Project Purpose
5. Research Question
6. Background on Semiconductor Technology
7. Background on Semiconductor Environmental Impacts
8. Literature Review
9. Summary of the Five Selected Research Papers
10. Data-Source Plan
11. Description of Selected Technological Features
12. Data Collection Methodology
13. Dashboard Architecture
14. Secure Dashboard Design
15. Threat Model
16. Attacker Scenario
17. Defender Strategy
18. Data-Integrity Controls
19. Incident-Response Process
20. Description of the Accurate Dashboard
21. Description of the Manipulated Dashboard

22. Side-by-Side Comparison
23. Analysis of Climate-Related Conclusions
24. Analysis of User Trust
25. Discussion
26. Limitations
27. Recommendations
28. Future Improvements
29. Conclusion
30. References

The report should explain how individual semiconductor technological features affect environmental or climate outcomes.

The report should also explain how manipulation of features, assumptions, sources, data, or visualizations could change the apparent climate impact.

The report should clearly distinguish between:

- Verified data
- Estimated data
- Assumptions
- Derived calculations
- Manipulated values used only for the adversarial demonstration

All academic sources should be properly cited, and the report should include a complete reference list.

2.6 In-Class Presentation or Demonstration

The team will give an in-class presentation or provide a video demonstration of the dashboard.

The presentation should summarize the major technical, environmental, and cybersecurity findings.

The presentation should include:

- Project purpose

- Research question
- Overview of semiconductor climate impacts
- Important semiconductor technology features
- Summary of the five research papers
- Description of the data sources
- Accurate dashboard demonstration
- Threat model
- Attacker scenario
- Defender strategy
- Manipulated dashboard demonstration
- Side-by-side comparison
- Incident-response process
- Security controls
- Major findings
- Limitations
- Final conclusions

During the demonstration, the team should clearly identify which dashboard is the trusted version and which dashboard represents the adversarial simulation.

The team should demonstrate how the same semiconductor climate data can lead to different conclusions when data, assumptions, sources, or visualizations are intentionally manipulated.

The presentation should conclude with the following major lesson:

> **Reliable climate analysis requires more than accurate numbers. The data source, assumptions, lifecycle boundaries, feature selection, security controls, and visualization design must also be protected from manipulation.**

3 Final Project Purpose

This project serves as the culminating assessment for the **Final Project Report & Presentation** grade.

The project is designed to demonstrate the team's ability to combine knowledge from multiple areas, including:

- Semiconductor technology
- Climate change
- Environmental data analysis
- Data visualization
- Cybersecurity
- Threat intelligence
- Threat modeling
- Data integrity
- Secure system design
- Incident response
- Adversarial analysis
- Research and academic communication

The final project should demonstrate not only how semiconductor technologies may affect climate change, but also how trustworthy climate conclusions depend on the security, integrity, provenance, assumptions, and presentation of the underlying data.

4 Weekly Project Deliverables

The project will be completed over four weeks. Each week will focus on a major stage of the project so that the research, cybersecurity analysis, dashboard development, and final presentation are completed in an organized sequence.

4.1 Week 1: Proposal, Research, and Data-Source Plan

During Week 1, the team will define the project scope, research question, semiconductor features, and data sources that will be used in the Climate-Feature Dashboard.

The main goal of Week 1 is to establish a strong research foundation for the project.

4.1.1 Week 1 Tasks

The team will:

- Define the overall project objective.
- Develop the main research question.
- Identify the semiconductor technology features that will be studied.
- Select at least five reliable academic or peer-reviewed papers.
- Read and summarize the five selected papers.
- Identify the environmental or climate-related features discussed in each paper.
- Determine what data will be used in the dashboard.
- Identify reliable data sources.
- Document the units, years, geographic scope, and lifecycle boundaries of the selected data.
- Identify limitations or uncertainties in the selected datasets.
- Create an initial data dictionary for the dashboard.

Possible semiconductor and climate features may include:

- Manufacturing energy consumption
- Electricity source
- Grid carbon intensity
- Water consumption
- Ultrapure-water use
- Greenhouse-gas emissions
- Fluorinated process gases
- Semiconductor technology node
- Manufacturing yield
- Chip materials

- Packaging materials
- Manufacturing location
- Supply-chain location
- Transportation impacts
- Waste generation
- Chemical use
- Lifecycle emissions
- Data-center energy use
- Chip operating efficiency

4.1.2 Paper Summary Requirements

For each of the five selected papers, the team should include:

- Full citation
- Research objective
- Semiconductor feature studied
- Data or variables used
- Climate or environmental impact discussed
- Main findings
- Limitations
- Explanation of how the paper will support the dashboard

4.1.3 Week 1 Deliverables

At the end of Week 1, the team should submit:

1. Project proposal
2. Research question
3. List of semiconductor climate features

4. Five selected academic papers
5. Summary of each paper
6. Data-source plan
7. Initial data dictionary
8. Preliminary reference list

4.2 Week 2: Threat Model, Data Integrity, and Incident-Response Plan

During Week 2, the team will focus on the cybersecurity portion of the project.

The main goal is to identify how the dashboard or its data could be manipulated and how defenders could detect, respond to, and recover from such an attack.

4.2.1 Week 2 Tasks

The team will:

- Identify the important assets that must be protected.
- Identify possible attackers.
- Identify attacker motivations.
- Create an attacker-versus-defender scenario.
- Identify possible attack vectors.
- Identify vulnerabilities in the dashboard environment.
- Identify possible data-integrity attacks.
- Determine how attackers could manipulate environmental data.
- Determine how attackers could manipulate dashboard assumptions.
- Determine how attackers could manipulate visualizations.
- Identify indicators of compromise.
- Design defensive security controls.
- Develop an incident-response plan.

4.2.2 Possible Attacker Scenarios

The threat model may include scenarios such as:

- An attacker steals an administrator password.
- A team member receives a phishing message.
- An attacker gains access to a server or cloud account.
- An attacker modifies a CSV or spreadsheet.
- An attacker changes values inside a database.
- An attacker modifies an API response.
- A malicious insider removes unfavorable climate data.
- A vendor provides incomplete or selectively reported environmental data.
- An attacker changes greenhouse-gas emission values.
- An attacker removes fluorinated process-gas emissions.
- An attacker modifies electricity carbon-intensity assumptions.
- An attacker removes geographic information related to water scarcity.
- An attacker changes chart scales to minimize the appearance of environmental impacts.
- An attacker replaces reliable sources with less reliable sources.

4.2.3 Security Controls

The team should consider controls such as:

- Multi-factor authentication
- Role-based access control
- Least-privilege access
- Encryption
- File hashing
- Checksums

- Digital signatures
- Version control
- Audit logs
- Secure backups
- Source verification
- Data validation
- Input validation
- Change approval
- Monitoring and alerting

4.2.4 Incident-Response Plan

The incident-response plan should include:

1. Preparation
2. Detection and Analysis
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

The team should explain how investigators would determine:

- What data were changed
- When the changes occurred
- Which user or account made the changes
- Which systems were affected
- Which dashboard features were affected
- Whether assumptions were changed

- Whether sources were removed
- Whether graphs or visualizations were manipulated
- How the trusted version of the dashboard would be restored

4.2.5 Week 2 Deliverables

At the end of Week 2, the team should submit:

1. Threat model
2. Attacker profile
3. Defender profile
4. Attacker-versus-defender scenario
5. Attack-vector analysis
6. Data-integrity plan
7. Security-control plan
8. Incident-response plan
9. Diagram of the dashboard data flow or system architecture

4.3 Week 3: Secure Dashboard Development and Manipulated Dashboard

During Week 3, the team will build the dashboard prototype and create the controlled adversarial version.

The main goal is to create both an accurate dashboard and a manipulated dashboard that can be compared directly.

4.3.1 Week 3 Tasks

The team will:

- Clean and organize the selected data.
- Standardize units where appropriate.
- Verify source information.

- Build the secure dashboard prototype.
- Add visualizations for selected semiconductor features.
- Add source information to the dashboard.
- Add assumptions and limitations.
- Add data-integrity information.
- Create a manipulated version of the dashboard.
- Document every manipulation made.
- Compare the accurate and manipulated versions.

4.3.2 Accurate Dashboard Components

The accurate dashboard may include:

- Energy-consumption charts
- Water-use charts
- Greenhouse-gas emission charts
- Process-node comparisons
- Packaging-material comparisons
- Manufacturing-location comparisons
- Renewable versus nonrenewable electricity comparisons
- Scope 1, Scope 2, and Scope 3 emission comparisons
- Carbon-intensity indicators
- Water-scarcity indicators
- Lifecycle-impact charts
- Supply-chain maps
- Historical trends
- Company or facility comparisons

The dashboard should also show:

- Data source
- Data year
- Measurement unit
- Geographic location
- Lifecycle boundary
- Assumptions
- Limitations
- Confidence or uncertainty

4.3.3 Manipulated Dashboard Components

The manipulated dashboard may intentionally include:

- Altered numerical values
- Deleted environmental features
- Hidden unfavorable information
- Changed assumptions
- Misleading chart scales
- Misleading colors
- Removed source information
- Selected favorable years only
- Selected favorable companies or facilities only
- Changed measurement units
- Hidden geographic information
- Removed water-scarcity information
- Excluded process-gas emissions

- Changed lifecycle boundaries
- Percentages instead of absolute values
- Intensity metrics instead of absolute emissions
- Cherry-picked favorable data
- Removed uncertainty
- Misleading labels or descriptions

4.3.4 Comparison Analysis

For each manipulation, the team should document:

- Original feature or value
- Manipulated feature or value
- Type of manipulation
- How the manipulation was performed
- Why a user may not notice the manipulation
- How the climate conclusion changes
- How user trust could be affected
- Which security control could detect the manipulation
- Which security control could prevent the manipulation
- What evidence would be useful during incident response

4.3.5 Week 3 Deliverables

At the end of Week 3, the team should submit:

1. Cleaned project dataset
2. Secure dashboard prototype
3. Accurate dashboard
4. Manipulated dashboard

5. Screenshots of both dashboard versions
6. Table documenting all manipulations
7. Initial comparison analysis
8. Dashboard security and integrity features

4.4 Week 4: Final Analysis, Written Report, and Presentation

During Week 4, the team will complete the project by analyzing the results, finalizing the written report, and preparing the presentation or recorded dashboard demonstration.

4.4.1 Week 4 Tasks

The team will:

- Complete the comparison between the accurate and manipulated dashboards.
- Analyze how manipulation changed climate-related conclusions.
- Analyze how manipulation could affect user trust.
- Evaluate the effectiveness of the proposed security controls.
- Review the incident-response process.
- Write the final project report.
- Review citations and references.
- Create presentation slides.
- Prepare the dashboard demonstration.
- Record or practice the final presentation.
- Complete final testing of the dashboard.

4.4.2 Final Written Report

The final report should include:

1. Title Page
2. Abstract or Executive Summary

3. Introduction
4. Project Purpose
5. Research Question
6. Background on Semiconductor Technology
7. Background on Semiconductor Environmental Impacts
8. Literature Review
9. Summary of Five Research Papers
10. Data-Source Plan
11. Description of Semiconductor Features
12. Data Collection Methodology
13. Data Validation Methodology
14. Dashboard Architecture
15. Secure Dashboard Design
16. Threat Model
17. Attacker Scenario
18. Defender Strategy
19. Data-Integrity Controls
20. Incident-Response Plan
21. Accurate Dashboard Analysis
22. Manipulated Dashboard Analysis
23. Side-by-Side Comparison
24. Climate-Impact Analysis
25. User-Trust Analysis
26. Discussion
27. Limitations

28. Recommendations
29. Future Improvements
30. Conclusion
31. References

4.4.3 Final Presentation or Demo

The final presentation should include:

- Project purpose
- Research question
- Semiconductor climate-impact overview
- Important semiconductor features
- Five research-paper summaries
- Data sources
- Accurate dashboard demonstration
- Threat model
- Attacker scenario
- Defender strategy
- Manipulated dashboard demonstration
- Side-by-side comparison
- Incident-response process
- Security controls
- Major findings
- Limitations
- Final conclusions

4.4.4 Week 4 Deliverables

At the end of Week 4, the team should submit:

1. Final secure dashboard
2. Final manipulated dashboard
3. Final accurate-versus-manipulated comparison
4. Completed written report
5. Complete reference list
6. Presentation slides
7. Recorded or live dashboard demonstration
8. Final project files and dataset

5 Four-Week Project Summary

Week	Focus	Main Deliverables
Week 1	Research and Data Planning	Proposal, research question, five papers, paper summaries, semiconductor features, data-source plan, and initial dataset.
Week 2	Cybersecurity and Threat Modeling	Threat model, attacker/defender scenario, data-integrity plan, security controls, incident-response plan, and architecture diagram.
Week 3	Dashboard Development and Adversarial Testing	Secure dashboard, manipulated dashboard, screenshots, manipulation documentation, and comparison analysis.
Week 4	Final Analysis and Presentation	Final report, final dashboards, comparison results, presentation slides, and dashboard demonstration.